The Relationship Between Classroom Air Quality and Student Performance

POSTED IN SCHOOLS.

Parents and teachers care deeply about our children's education. Who doesn't want them to have the most fruitful experience possible? One aspect of education that is often neglected is indoor air quality. Good air quality in schools can improve student performance significantly and it deserves as much attention as any item on a school planner's agenda. Attendance, productivity, and other performance indicators are shaped by it. The consensus is clear — students and staff need clean air.

Indoor air contaminants are well-known causes of absences, interruptions, and lack of productivity in students, faculty, and staff. In a poll of Chicago teachers, 25% said respiratory problems, including asthma, are the most encountered problems relating to school facility quality. Another 16% reported problems that are often caused by poor indoor air quality (IAQ), such as sinus infections.[1] The EPA states that indoor air quality can help to improve test scores and attendance,[10] and science has given us insight as to why that is.

The Effects of Poor Air Quality in Schools

Supporting students and faculty by providing high-quality air can make a difference. Studies have shown that student performance improves as classroom air quality improves. In such studies, the relationship between student performance and good air quality explains the positive correlation between them. Air quality has a direct impact on students but also on their teachers and others involved in their school experience.

Asthma

There has been little research into asthma symptoms caused by school air quality. However, a 2013 study found that mouse allergen was significantly more abundant in schools than in students' homes and that higher levels of mouse allergen contributed to a 27% increase in asthma symptom occurrence.[6] Another study found that schools with new ventilation systems installed reported a reduction in asthma symptoms.[7]

Mouse allergen is not the only allergen found is schools, of course. Some of the most common asthma triggers include tobacco smoke, dust mites, mold, cockroaches, and other pests,[5] all of which must be mitigated in schools.

According to the EPA, 12.8 million school days were lost due to asthma attacks in 2003.[3] Though it's unclear how many of those resulted from school air quality (versus home and outdoor air quality), absenteeism could have been reduced if indoor air triggers were not present in schools.

Teacher Substitution

Teachers experience illness caused by poor indoor air quality just as students do. Sometimes this leads to teacher absence. The disruption caused by substituting a class's permanent teacher cannot be fully compensated for and has a negative impact on student achievement. A study of 62 Florida schools found that students who spent four weeks or more with a substitute teacher scored 11 points lower on the reading portion of the Florida Comprehensive Assessment Test.[4]

Student Absenteeism

22% of absences in a study of North Carolina schools were caused by respiratory illnesses such as asthma and allergies.[8] In a review of 11 studies, 7 of them found an inverse correlation between absence rate and school performance.[9] That is, a student's performance is usually poorer as the student's absence rate increases.

Additional Concerns

Poor air quality in schools can bring on numerous health concerns that reduce student performance, ranging from nasal congestion to respiratory infection. Even school staff members who don't regularly interact with students can have a negative impact on student performance when poor air quality has affected their health. Staff may spread contagious illness within the school facilities. Their work performance may decline as well, which can affect students in various ways. Everyone involved needs high air quality standards.

Good Indoor Air Quality is a Necessity in Schools

Many scientists and statisticians have delved into the question of what factors can affect student and faculty performance. Factors that have been researched and tested include temperature, air quality, lighting, acoustics, and class size. Many studies place a high level of importance on air quality.

A 2004 paper [2] consolidates empirical studies from the 1970s until the early 2000s to rank 31 criteria in terms of their effect on "school building adequacy." Glen Earthman, the author, is an experienced school facility planner and former director of the Educational Resource Information Center (ERIC). Dr. Earthman found significant evidence that indoor air quality is one of the most important factors in student learning. In his prioritization, the only factor that ranks higher than air quality is "human comfort" (i.e. air temperature). Earthman describes indoor air quality as "appropriate ventilation and filtering systems, as regulated by appropriate HVAC systems."

Bipolar Ionization Boosts Air Quality and School Performance

You may be asking what you can do to improve the air quality in your facility. One of the most effective solutions is called bipolar ionization, an air purification technology that reduces a wide range of air contaminants. Cleaning the air with a high-quality ionization device reduces and often prevents the effects of asthma, allergies, and other illnesses, including influenza A.

Getting Started on Improving School IAQ

While air quality is not the only determinant of strong grades and future success, it is indeed a factor worth consideration for responsible leaders in the education industry. While the technology on the cutting edge of air quality sounds complicated, it often does not require a huge investment.

Air purification is an adaptable solution, as well. Plasma Air HVAC products have been deployed in new school facilities and retrofitted into existing HVAC systems. Our medical-grade portable units are an ideal solution designed for rapid remediation in situations where infection risk is high, complimenting surface cleaning and hand hygiene protocols.

We aim to provide a simple, cost-effective solution. Get in touch with us to learn more.

Sources

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